Bhusan Shrestha

DevOps Engineer | Aspiring Security Engineer

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Summary

DevOps Engineer experienced in deploying and managing applications in Linux-based environments, configuring web infrastructure, automating deployment workflows, and troubleshooting infrastructure and application issues. Works with Docker, Docker Swarm, Kubernetes, Nginx as reverse proxy and load balancer, Cloudflare, SSL/TLS and CI/CD. Strong understanding of software development and backend systems (Node.js, Spring Boot, PostgreSQL, Redis), with a focus on building reliable, secure and scalable deployment environments.

Profiles

LinkedIn
bhusan-shrestha

GitHub
Bhusan-Shrestha

Experience

PortPro Technologies, Inc. Pattipa Marg Uttar, Lalitpur
DevOps Intern August 2026 – November 2026

- Independently built a todo application (frontend, backend API and a todo-stats microservice) and took it from code to a secured, monitored Docker Swarm deployment on AWS, with Terraform-provisioned servers and automated pipelines with rollback.
- Wrote Terraform code to provision AWS EC2 servers: two instances share one security group with a self-referencing rule, so they communicate over private IPs without opening an IP range to the internet, and SSH is restricted to a single admin IP. The application runs on these servers behind Cloudflare with proxying enabled.
- Built GitHub Actions CI/CD pipelines: one automatically runs tests, builds the Docker images, pushes them to a registry and deploys the stack to the Swarm, and a separate rollback pipeline restores a previous version when a release needs to be reverted.
- Controlled service placement across manager and worker nodes: the backend runs on the manager and publishes host port 4000, while todo-stats runs on worker nodes only, spread across them, reachable only over an internal overlay network. Nginx sits in front as the reverse proxy.
- Tested the same application on Kubernetes using k3s instead of Docker Swarm, running the backend and todo-stats services as Kubernetes workloads in a k3s cluster. Created YAML manifests for Ingress, Namespace, Deployment, Service, ConfigMap, and Secret resources.
- Hardened the containers: the backend image creates a dedicated unprivileged user and runs as it, and the stats image runs as the built-in non-root node user, so neither application container runs as root. Both use minimal Alpine-based Node images.
- Reduced exposure and set limits: only the backend publishes a port, both services have CPU and memory limits and restart policies, and todo-stats has a health check.
- Secured PostgreSQL on the same server: it accepts connections only from the Docker Swarm IPs, and only to the tododb database through a dedicated low-privilege todo user, so nothing else can reach the database.
- Set up monitoring and observability with OpenTelemetry, Prometheus, Grafana, Tempo and Loki, covering metrics, traces and logs.
- Wrote Bash scripts to automate setup tasks, including the Nginx, Docker and PostgreSQL setup.
- Wrote a deployment guide with the build, push and deploy steps and troubleshooting notes for common deployment errors.

https://portpro.io/

Education

Himalayan Whitehouse International College — BIT (Bachelor in Information Technology)
2022 – 2026

Seabird International Secondary School — +2 Level
2020 – 2022

Projects

Todo Application - Docker Swarm Deployment

Todo application deployed as Docker Swarm microservices behind an Nginx reverse proxy, with GitHub Actions pipelines that test, build, push, deploy and roll back releases, plus Bash automation scripts and observability through OpenTelemetry, Prometheus, Grafana, Tempo and Loki. Containers run as non-root users, only the backend publishes a port, and services have CPU and memory limits. PostgreSQL runs on the same server and accepts connections only from the Swarm IPs, through a low-privilege todo user limited to the tododb database. Also tested on Kubernetes with k3s in place of Swarm.

https://github.com/Bhusan-Shrestha/todo-app

AWS EC2 Infrastructure with Terraform

Provisioned two AWS EC2 instances using Terraform, configuring both instances to share a security group with self-referencing rules to enable secure communication over private IP addresses without exposing internal traffic to the internet. Restricted SSH access to a single authorized administrator IP to enhance security. Deployed the application on the EC2 instances behind Cloudflare with proxying enabled for an additional layer of protection and traffic management.

https://github.com/Bhusan-Shrestha/Terraform-EC2

Movie Streaming Platform

Developed a full-stack movie streaming website using React.js for the frontend and Java Spring Boot for the backend REST API. Implemented movie search and pagination features to help users efficiently discover movies and navigate through large collections of content. Integrated the frontend with REST API endpoints to retrieve and display movie data, gaining hands-on experience in full-stack development, API integration, and frontend-backend communication.

https://github.com/Bhusan-Shrestha/Movie-Streaming-Website

Business Sales Prediction

Developed a full-stack business intelligence (BI) system for sales and inventory analysis, providing insights into business performance and inventory trends. Integrated an XGBoost machine learning model to forecast future sales and support data-driven decision-making. Implemented an AI-powered chatbot to enable interactive data queries and help users access business insights more efficiently.

https://github.com/Bhusan-Shrestha/Power-BI-using-XGBoost

Technical Skills

DevOps & Infrastructure

Linux, Docker, Docker Compose, Docker Swarm, Kubernetes (k3s), Terraform, Ansible, Nginx, Git/GitHub, CI/CD (GitHub Actions), Bash, Networking, VPS deployment

Monitoring & Observability

Grafana, Prometheus, Loki, Tempo, OpenTelemetry

Cloud & Deployment

AWS, Cloudflare, Containerized deployment, Container security basics, Load balancing, Reverse proxy, SSL/TLS

Backend & Development

Node.js, Express, NestJS, Java Spring Boot, PHP, REST APIs

Databases

PostgreSQL, MySQL, MongoDB, Redis

Frontend

React.js, Next.js, JavaScript, HTML, CSS, Tailwind CSS

Soft Skills

Troubleshooting & Problem-Solving

Finds the cause of infrastructure and application problems, from failed deployments and container errors to network and proxy issues.

Communication & Team Collaboration

Explains technical work clearly and works with teammates on shared projects.

Self-Directed Learning

Learns new tools independently and puts them to use in working projects, such as the Docker Swarm and monitoring stack.

Attention to Detail

Careful with configuration, testing and debugging.

Badge

AWS Academy Graduate - Cloud Foundations - Training Badge — September 2026

Amazon Web Services Training and Certification

https://www.credly.com/badges/c8b2168d-2b8e-4ae1-a07e-2038bc4ad052/linked_in_profile

Certifications

Network Security

Great Learning

https://greatlearning.in/academy/learn-for-free/courses/network-security

Cybersecurity Architecture Fundamentals and Tools

Mind Luster

https://www.mindluster.com/

Encryption Basics

Great Learning

https://greatlearning.in/academy/learn-for-free/courses/encryption

Introduction To Data Science Using Python

Himalayan Whitehouse International College

https://whitehousecmt.edu.np/

Data Analysis And Data Science Techniques

AITC International Pvt. Ltd.

https://aitc.ai/

Awards

Outstanding Design Skills — February 2025

Whitehouse Innovator

Awarded by Whitehouse Innovator for outstanding design skills at the Frontend Design Fest.

2nd Position, IT Fest 2081 — March 2025

ANNFSU Himalayan Whitehouse International College Committee

Second position at IT Fest 2081, held on 24 and 25 March 2025 at Himalayan Whitehouse International College.

Interests

DevOps

Cyber Security

Artificial Intelligence

Emerging Technologies

Languages

English | **Nepali** | **Newari**